02 · Lesson spec

Let's Build It! · 40 minutes · whole class, then groups of 4–6 · fully unplugged

Goal	**I can build a bridge or ramp and test it.**
Core parts	Move + Build
Key words	bridge · ramp · tower · strong · test · push · pull · roll
Materials	Blue paper river per group; big blocks, boxes, card; a plank or ramp board; a large toy car or plastic bottle (wheels checked); a ball; a box on a short string; build card and plan cards (Printables)



1. Watch (5 min)

- **Show** the blue paper river and Ayo's toy car. **Ask:** "The car must cross the river. How can we help it?" Take ideas: a bridge, a ramp.
- **Show the plan cards:** blocks → plank → test. "Engineers **plan**, then **build**, then **test**. If it doesn't work, we try again!"
- **Hand signs:** build = fists stacking · bridge = arms make an arch · ramp = slanted arm · strong = show muscles · test = thumbs up/down wiggle · push = both palms forward · pull = hands pull a rope · roll = roll hands forward.

2. Plan, build, test (25 min)

a) Push, pull, roll (8 min) · Core: Move

- **Push** a large toy car across the floor. **Pull** a box on a short string. **Roll** a ball or a large plastic bottle down a low ramp, then a higher ramp.
- **Ask:** "Which ramp made it go further?" Cheer when it reaches the end!

b) Build a bridge or ramp, groups (12 min) · Core: Build

- **Code:** each group lays the 3 plan cards (blocks → plank → test) and follows the plan.
- Build a bridge over the river (blocks on each side, a plank on top) or a ramp down to the river bank. **Test** it with the toy car or bottle.
- **If it falls or the car can't cross:** "Try another way!" (W4). Change one thing (a bigger block, a wider base, a lower ramp), then test again.
- The teacher **photos** each tested build (evidence).
- **Stretch:** make the bridge hold two cars, or build a tower as tall as your knee that stands.

c) Activity Book page (5 min)

- Circle the strong bridge; match push and pull pictures. Not marked.

3. Share (10 min)

- **Create:** decorate your bridge and give it a name ("Super Bridge", "Ayo Bridge").
- Each group tests its bridge in front of the class: "Will the car cross?" Countdown and cheer.
- **Close:** "Plan, build, test, and if it falls, try again!"

Common mistakes

- Building without a plan → "Show me your plan cards first."
- Giving up when it falls → "Engineers try again! What can we change?"

- Changing everything at once → "Change one thing, then test."

What if it doesn't work?

- No plank → a strip of strong card or a book. No toy car → a large plastic bottle rolls well.
- Not enough blocks → shoeboxes and cereal boxes.

Support and stretch

- **Support:** build the bridge pillars together; the child places the plank and tests.
- **Stretch** (optional, never assessed): hold two cars, or a knee-high tower that stands.

How we check

Observation only. Tick each child who **builds a bridge or ramp and tests it with a toy car, and uses push, pull or roll**. Take a photo.

> **Safety**
> Adults do any cutting; big pieces only. Wheels checked. No climbing on builds. Short string, never round a neck. Towers no taller than a knee.